WORKED SOLUTION

STEP II 2018, Question 1

Pure • Roots of Polynomials

STEP II 2018, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q1

The key idea

The coefficients read the same forwards and backwards, so dividing by x^2 turns the quartic into a quadratic in $t = x + \frac{1}{x}$. Each value of t gives a pair of roots k, $\frac{1}{k}$ of the quartic, and a real x can only give $|t| \geq 2$, with $t = \pm 2$ exactly when $x = \pm 1$. Counting distinct roots then comes down to asking which values of t occur and whether either of them is ± 2.

Solution

Roots come in reciprocal pairs

Write $P(x) = x^4 + ax^3 + bx^2 + ax + 1$. Since $P(0) = 1 \neq 0$, any root k is non-zero, and

$$k^4 P\left(\tfrac{1}{k}\right) = 1 + ak + bk^2 + ak^3 + k^4 = P(k) = 0.$$

As $k^4 \neq 0$, this gives $P(\frac{1}{k}) = 0$, so k^{-1} is a root.

The substitution $t = x + \frac{1}{x}$

Dividing $P(x) = 0$ by x^2 (allowed, as $x \neq 0$) and using $x^2 + \frac{1}{x^2} = t^2 - 2$:

$$\left(x^2 + \frac{1}{x^2}\right) + a\left(x + \frac{1}{x}\right) + b = 0 \iff t^2 + at + (b - 2) = 0.$$

Let t_1, t_2 be the roots of this quadratic, so $t_1 + t_2 = -a$ and $t_1 t_2 = b - 2$. Expanding shows

$$P(x) = (x^2 - t_1 x + 1)(x^2 - t_2 x + 1),$$

because the right-hand side is $x^4 - (t_1 + t_2)x^3 + (2 + t_1 t_2)x^2 - (t_1 + t_2)x + 1$. So the roots of P are the roots of $x^2 - t_1 x + 1$ together with those of $x^2 - t_2 x + 1$.

Now use the assumption that all the roots are real. Each factor $x^2 - t_i x + 1$ has two real roots k and $\frac{1}{k}$ (their product is 1), and $t_i = k + \frac{1}{k}$ is real. For real $k \neq 0$, $k + \frac{1}{k}$ has the

sign of k, and

$$\left|k+\tfrac{1}{k}\right|-2=\frac{(|k|-1)^2}{|k|}\geq 0,$$

with equality only when $k=\pm 1$. So:

- if $|t_i|>2$, the factor x^2-t_ix+1 has two distinct real roots, neither equal to ± 1;
- if $t_i=2$, the factor is $(x-1)^2$; if $t_i=-2$, it is $(x+1)^2$.

Also, different values of t cannot share a root x, since x determines $t=x+\frac{1}{x}$.

Part (i)

If there is only one distinct root k, then $\frac{1}{k}=k$, so $k=\pm 1$ and $P(x)=(x\mp 1)^4$. Expanding,

$$(x-1)^4=x^4-4x^3+6x^2-4x+1,\qquad (x+1)^4=x^4+4x^3+6x^2+4x+1,$$

both of the required form. So the values are $a=-4,\ b=6$ and $a=4,\ b=6$.

Part (ii)

Run through the possibilities for t_1, t_2 (each real with $|t_i|\geq 2$):

- $t_1\neq t_2$, both with $|t_i|>2$: four distinct roots.
- $t_1=t_2$: the quartic is a perfect square $(x^2-t_1x+1)^2$, so it has two distinct roots if $|t_1|>2$ and one if $t_1=\pm 2$.
- $\{t_1,t_2\}=\{2,-2\}$: $P(x)=(x-1)^2(x+1)^2$, two distinct roots.
- One of the t_i is ± 2 and the other has modulus greater than 2: roots ± 1 (double) and two others, which is three distinct roots.

Ask yourself Why must one of the values of t be ± 2?

Three distinct roots out of four means some root is repeated, and the list above shows that this happens with three distinct roots only in the last case. Put more directly: the roots split into reciprocal pairs, and an odd number of distinct roots forces one root to be its own reciprocal, i.e. ± 1.

So $t=2$ or $t=-2$ is a root of $t^2+at+b-2=0$:

$$\begin{aligned} t=2:&\quad 4+2a+b-2=0 \iff b=-2a-2,\\ t=-2:&\quad 4-2a+b-2=0 \iff b=2a-2.\end{aligned}$$

Hence either $b=2a-2$ or $b=-2a-2$, as required.

Part (iii)

With $b=2a-2$ the quadratic in t is

$$t^2+at+2a-4=(t+2)(t+a-2),$$

so $t_1 = -2$ and $t_2 = 2 - a$. The first gives $(x+1)^2$, and the second gives $x^2 - (2-a)x + 1 = 0$, so

$$x = -1 \text{ (twice)}, \qquad x = \frac{2 - a \pm \sqrt{a^2 - 4a}}{2}.$$

These are real because we are told all roots are real; this amounts to $a^2 - 4a = a(a-4) \geq 0$, i.e. $a \leq 0$ or $a \geq 4$.

The final part

By (ii), three distinct roots needs $b = 2a - 2$ or $b = -2a - 2$. Conversely, suppose $b = 2a - 2$ (with all roots real). From (iii) the values of t are -2 and $2 - a$, and from the case list we get exactly three distinct roots unless $2 - a = \pm 2$:

- $a = 0$, $b = -2$: $t = \pm 2$, $P(x) = (x^2 - 1)^2$, only two distinct roots;
- $a = 4$, $b = 6$: $t = -2$ twice, $P(x) = (x+1)^4$, one root.

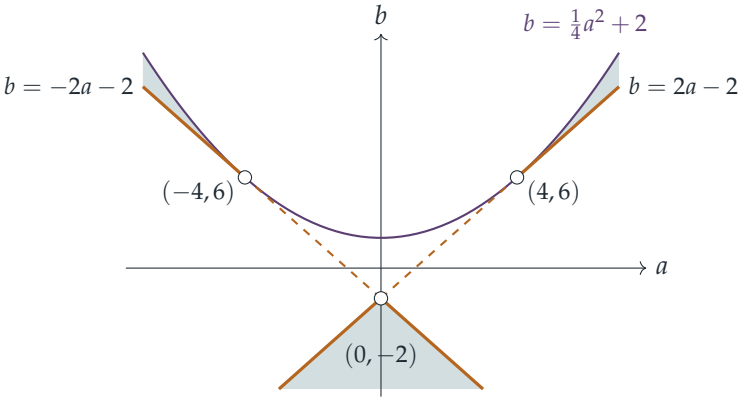
Replacing x by $-x$ turns P into the same form with a replaced by $-a$, which swaps the two lines. So for $b = -2a - 2$ the values of t are 2 and $-a - 2$, and the exceptions are $a = 0$, $b = -2$ and $a = -4$, $b = 6$.

Therefore, for real a, b with all roots real, $(*)$ has exactly three distinct roots if and only if

$$b = 2a - 2 \text{ or } b = -2a - 2, \qquad \text{with } (a,b) \neq (0,-2),\ (4,6),\ (-4,6).$$

Note that $(4,-10)$ and $(-4,-10)$ are allowed: for example $a = 4$, $b = -10$ gives $t = 2$ and $t = -6$, so three distinct roots.

The picture shows the (a,b)-plane. The shaded region is where all roots are real: below both lines (one value of t is at most -2 and the other at least 2), or between a line and the parabola $b = \frac{1}{4}a^2 + 2$ for $|a| \geq 4$ (both values of t on the same side). The parabola is where $t_1 = t_2$, and it touches the lines at $(\pm 4, 6)$. The answer is the solid parts of the two lines on the edge of the shaded region, minus the three open circles.



(The dashed parts of the lines, $0 < |a| < 4$, lie outside the region where all roots are real, which is why (iii) needed $a \leq 0$ or $a \geq 4$.)

Checks

- Take $a = -4$, $b = -10$ (on $b = 2a - 2$, allowed). Then (iii) gives $x = -1, -1, 3 \pm 2\sqrt{2}$, and indeed $(x+1)^2(x^2 - 6x + 1) = x^4 - 4x^3 - 10x^2 - 4x + 1$. The roots $3 \pm 2\sqrt{2} \approx 5.828,\ 0.172$ are reciprocals, as they should be.
- At the excluded point $a = 0$, $b = -2$ the quartic is $x^4 - 2x^2 + 1 = (x^2 - 1)^2$, with only two distinct roots.
- A computer found the roots numerically for every (a, b) on a grid of step $\frac{1}{4}$ with $|a| \leq 10$, $|b| \leq 15$; of the 1579 grid points with all roots real, exactly the 88 predicted by the condition above gave three distinct roots.

Common slips

- Forgetting the exceptional points: $b = 2a - 2$ alone is necessary but not sufficient, since $(0, -2)$ and $(4, 6)$ lie on it.
- Assuming that a repeated root must be ± 1: a double pair $k, k, \frac{1}{k}, \frac{1}{k}$ is also possible, but it gives two distinct roots, not three.

Beyond STEP

Polynomials whose coefficients read the same both ways are called *palindromic* (or reciprocal), and the trick $t = x + \frac{1}{x}$ halves the degree of any of even degree. The same substitution with $x = e^{i\theta}$ gives $t = 2\cos\theta$, which is how the minimal polynomials of $2\cos(2\pi/n)$ are found and how Gauss handled the regular 17-gon. The idea of describing where a polynomial's roots collide, as the curve in the (a, b)-plane above does, leads to the *discriminant*. Allenby's *Rings, Fields and Groups* is a good place to read more about polynomials and their roots.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.